IPS Academy, Indore
Institute of Engineering & Science · Dept. of Computer Science & Information Technology
Session 2026-27

LeadGen

AI-Powered Lead Management System



GUIDED BY
Mrs. Poonam Mishra

SUBMITTED BY · GROUP NO. 52
Trisha Mishra (0808CI231214)
Ratnesh Ranve (0808CI231162)

Problem, Objectives & Expected Outcome

Problem Statement

Leads come from websites, emails, ad platforms and other apps. When they are handled by hand, the same customer gets saved twice, follow-ups are late and nobody knows which lead to call first, so good customers are lost.

Objectives

1. Collect leads from every source in one place and catch duplicates
2. Score each lead with ML as High, Medium or Low
3. Give each lead to the right salesperson automatically
4. Give teams dashboards, reports and AI-drafted replies

Expected Outcome

- Faster first response to every new lead
- No two salespeople calling the same customer
- Time spent on the leads most likely to buy
- Decisions based on live reports

Research gap addressed: earlier studies look at lead scoring, lead handling and collecting leads separately. LeadGen does all of it in one automatic flow (collect → duplicate check → ML score → assign → AI-assisted reply) and the model keeps learning from results.

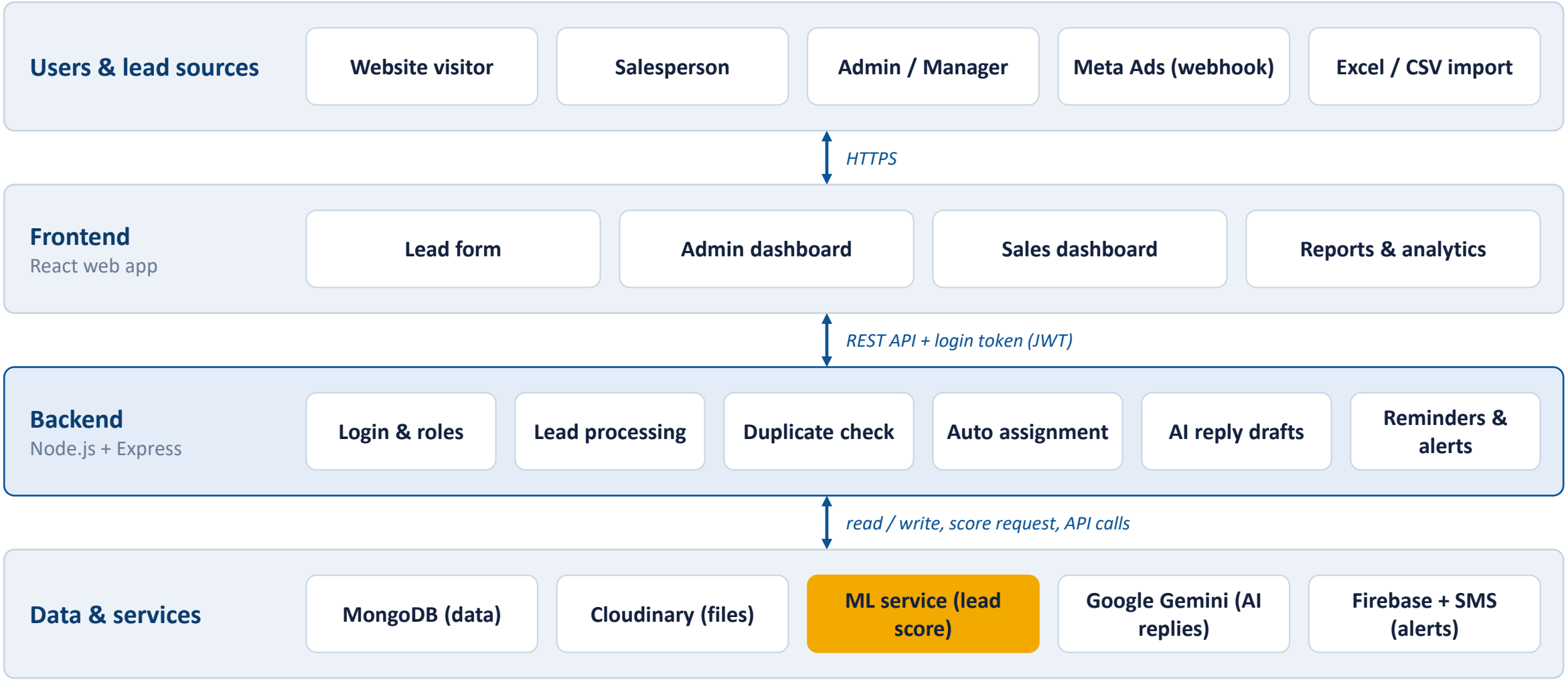
Literature Review Matrix

#	Study (authors, title, venue, year)	Method	Key contribution	Gap LeadGen fills
1	Espadinha-Cruz, Fernandes & Grilo, "Lead management optimization using data mining: A case in the telecommunications sector," Computers & Industrial Engineering (Elsevier), 2021	Data mining to sort leads	Predicts which leads will convert	No automatic handling; one lead source only
2	Ohiomah et al., "The role of lead management systems in inside sales performance," Journal of Business Research (Elsevier), 2019	Survey of 108 sales staff	Lead systems improve follow-ups and selling	No AI scoring; no automatic assignment
3	Das & Datta, "Application of UTAUT2 on adopting AI-powered lead management system (AI-LMS) in passenger car sales," Technological Forecasting and Social Change (Elsevier), 2024	Survey on adopting AI tools	Why sales teams accept AI lead tools	Studies adoption; no system built
4	Nygård & Mezei, "Automating lead scoring with machine learning: An experimental study," Proc. HICSS, 2020	ML models on CRM data	ML lead scoring from past results	Scoring only; no collection, assignment or follow-up
5	Wu, Andreev & Benyoucef, "The state of lead scoring models and their impact on sales performance," Information Technology and Management (Springer), 2024	Review of lead scoring models	Compares scoring methods and their effect on sales	Review only; no working system

Common gap: *no single system collects leads from many sources, removes duplicates, scores with ML, assigns automatically and shows reports.*

System Architecture

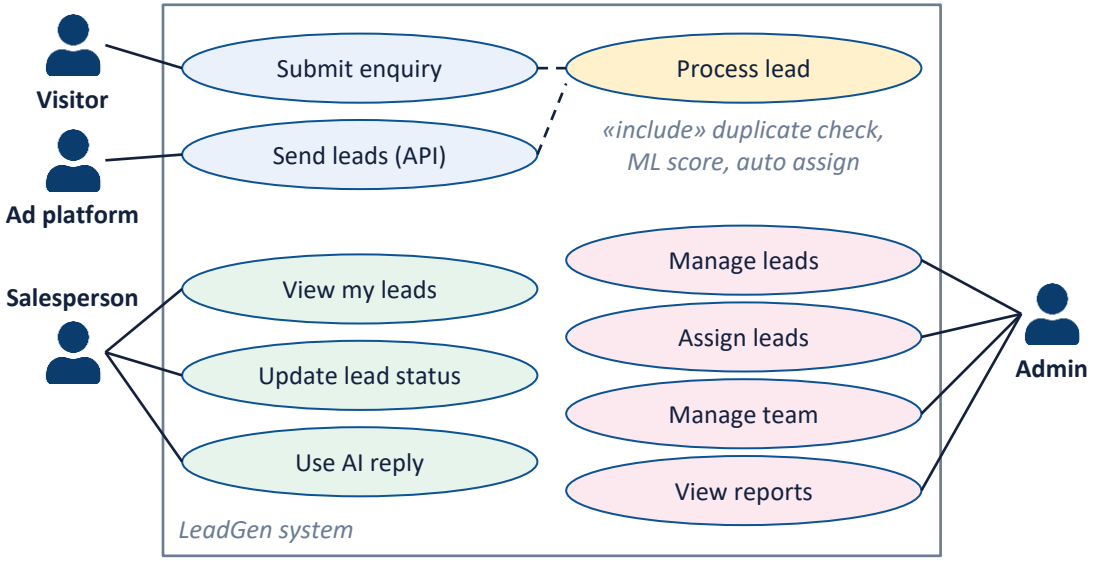
[UPDATED]



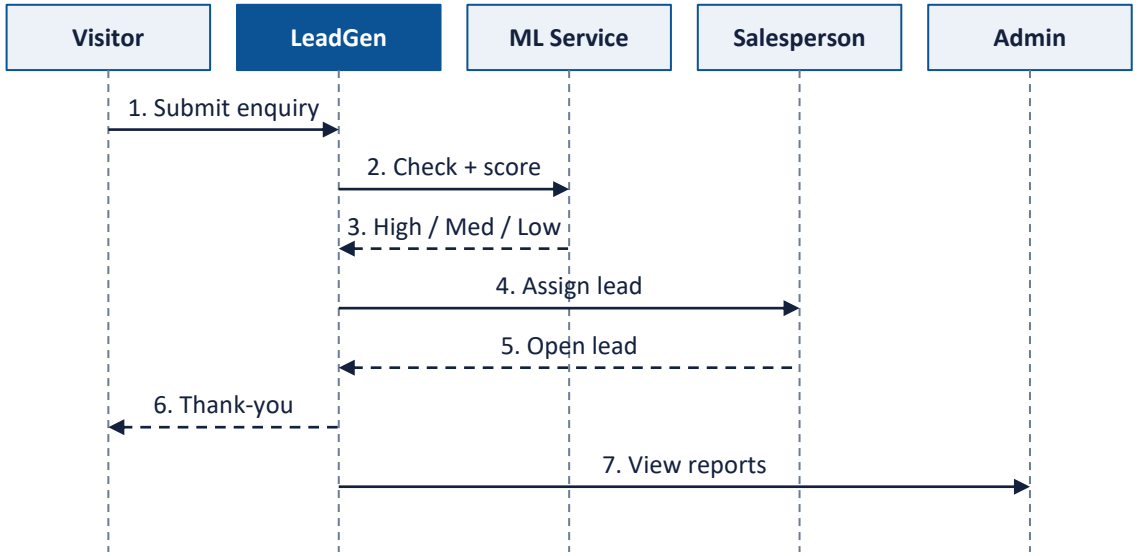
UML System Design

[UPDATED]

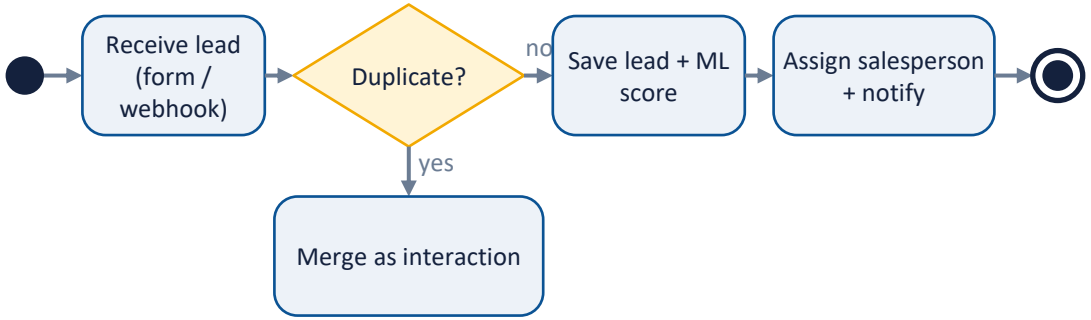
Use Case Diagram



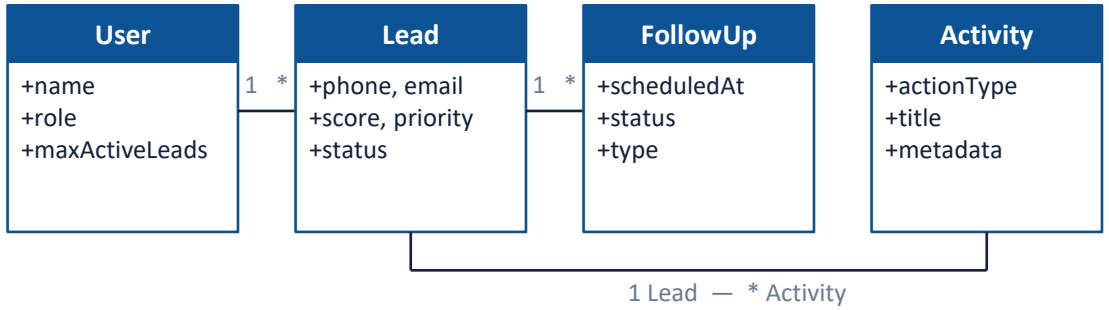
Sequence Diagram



Activity Diagram: lead intake

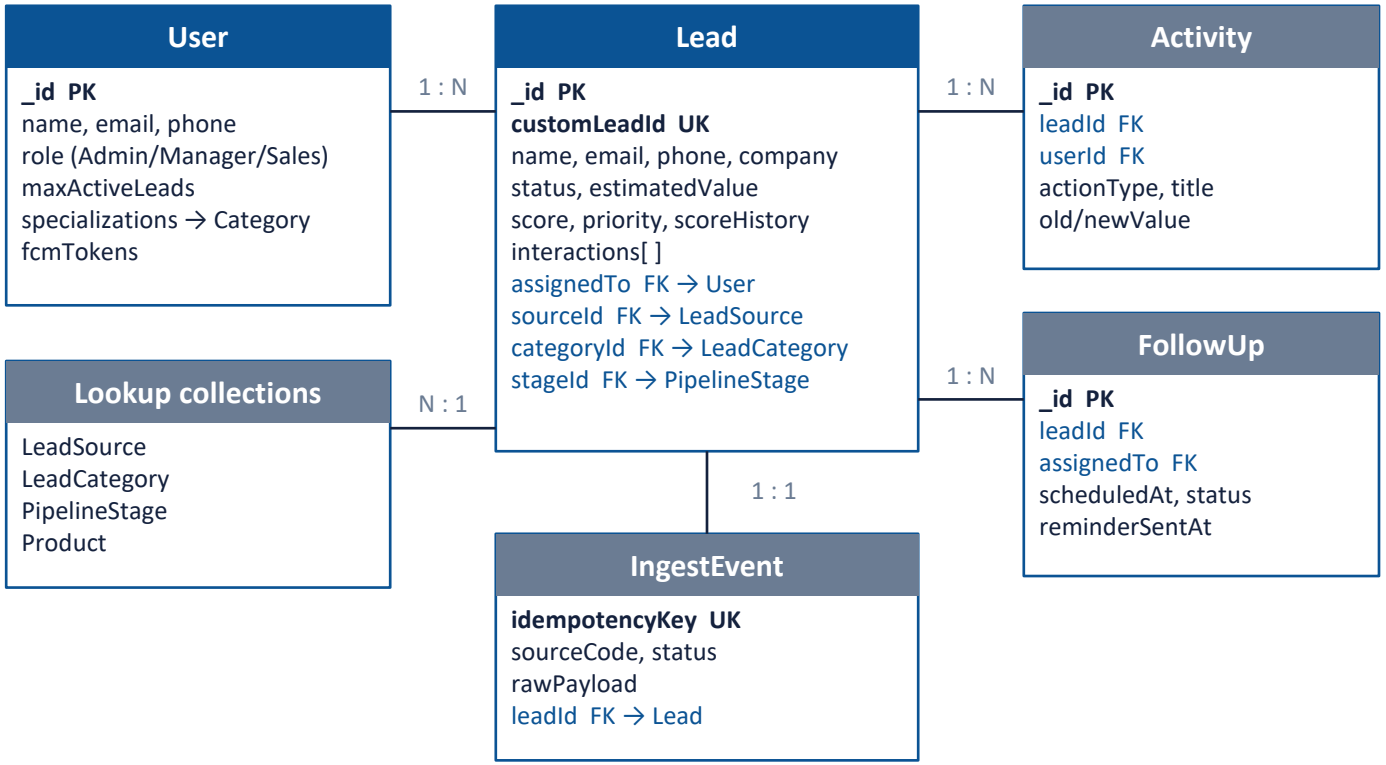


Class Diagram: core domain



Database Schema & API Contracts

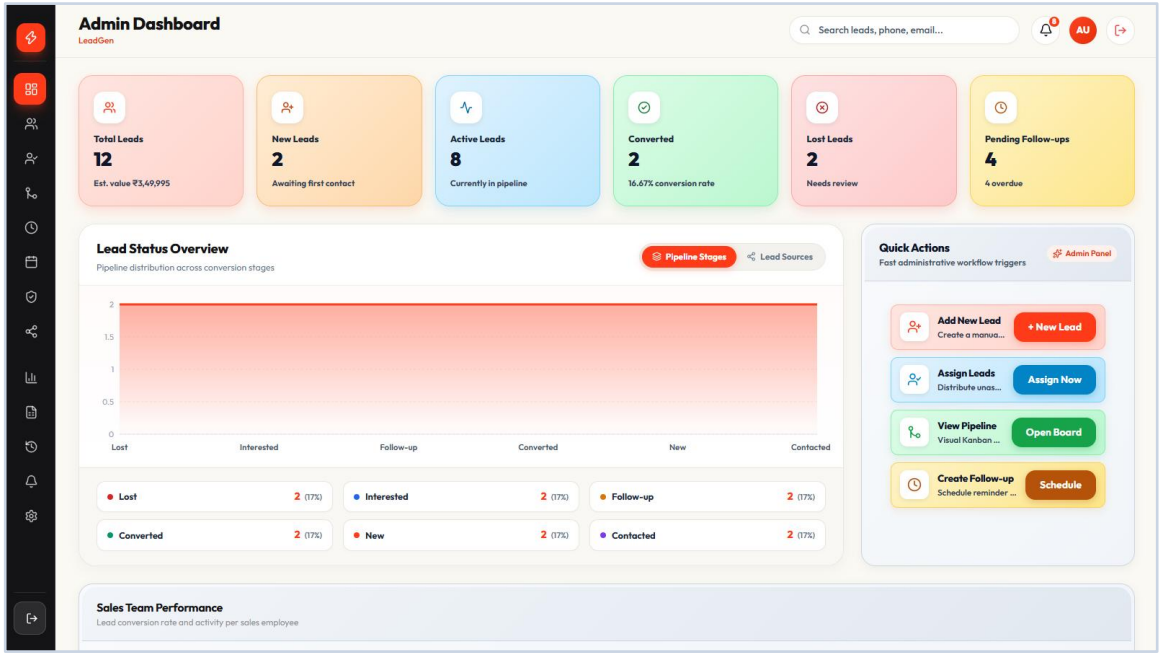
[UPDATED]



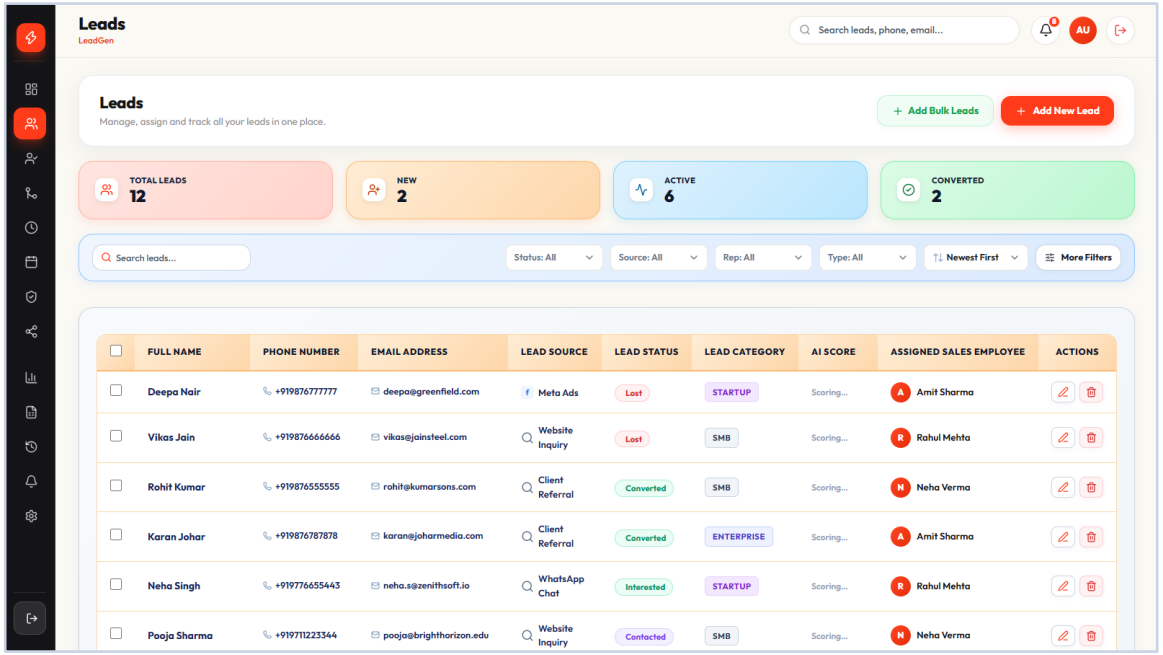
Method	Endpoint (/api/v1)	Who can use it → what it does
POST	/auth/login	Anyone → login tokens (15 min, renewed for 7 days)
POST	/ingest/public	Anyone, limited per IP → website form lead
POST	/ingest/webhook/:sourceCode	Ad platform, signed → same request saved once
GET	/leads	Logged-in user, by role → list & filter leads
POST	/leads/:id/interactions	Logged-in → log a call/reply, re-score lead
POST	/leads/:id/ai-draft	Logged-in → Gemini writes a reply draft
GET	/dashboard/stats, /reports/*	Admin / Manager → numbers & reports
POST	ML /predict	Backend only (API key) → score + High/Med/Low

MongoDB Atlas · Mongoose 8 · 17 collections in total (also Task, Notification, AuditLog, OtpVerification, Setting, CustomField, RolePermission)

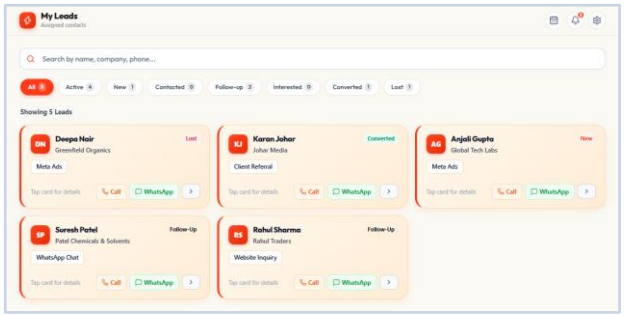
Phase-1 Code Screenshots



Admin dashboard: lead counts and pipeline chart



Leads table: source, status, AI score, assigned salesperson



Sales panel: My Leads

```
// leadAssignment.service.js: give the lead to the least busy salesperson
let eligible = reps
  .map((rep) => ({ rep, workload: workloadMap.get(rep._id.toString()) || 0 }))
  .filter(({ rep, workload }) => workload < (rep.maxActiveLeads || 20));
// prefer salespeople who specialise in this lead's category
eligible.sort((a, b) => a.workload - b.workload);
await leadService.assignLead(leadId, eligible[0].rep._id, systemUser);
```

Code: automatic lead assignment

Team Work Division & Timeline

TM

Trisha Mishra

0808CI231214

- Frontend: React web app, admin and sales panels
- ML model training (together)

RR

Ratnesh Ranve

0808CI231162

- Backend: Node.js APIs, database, lead flow
- ML model training (together)

Project timeline



Requirement & Planning

Workflow, features, tech stack

August 2026

Phase-1 Development

React UI, Node APIs, MongoDB, ~50% code

September 2026

Phase-2 Final Sprint

ML scoring, auto assignment, AI replies, testing

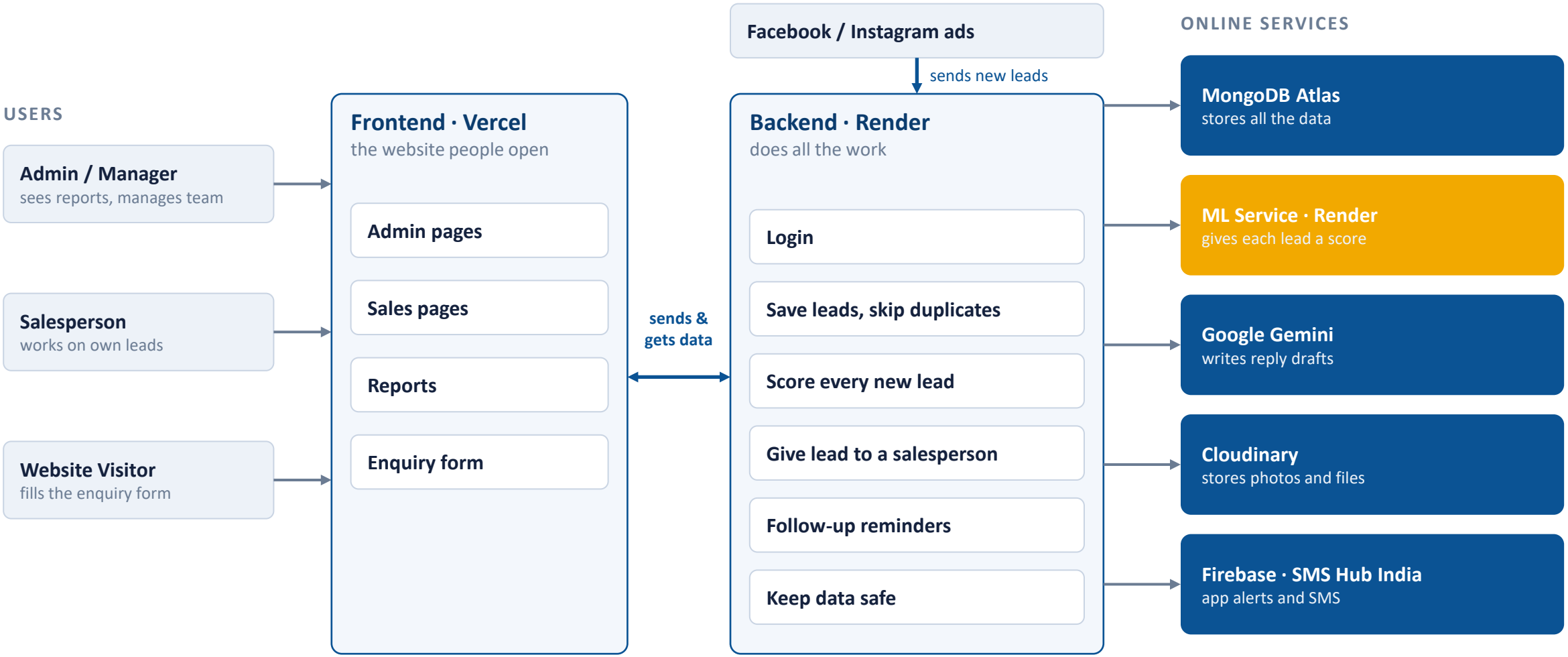
September 2026

Deployment & Defense

Vercel + Render + Atlas, final viva

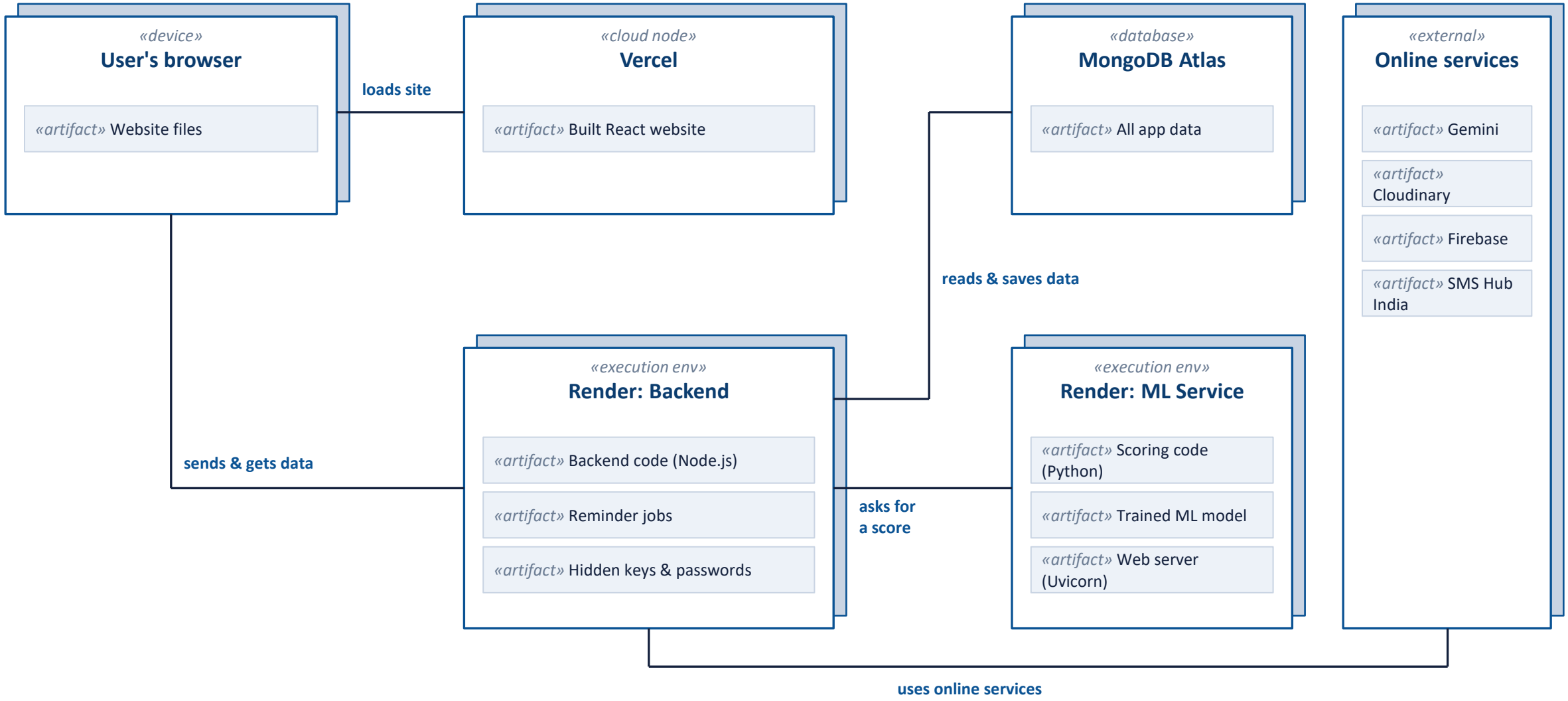
October 2026 (by 6th)

Final Deployed Architecture & Cloud Integration



How a lead moves: form or ad → duplicate check → saved → scored → given to a salesperson → alert sent

UML Deployment Diagram



100% Working Live System Demo

1. **Admin login**
   Dashboard with live numbers and charts

2. **Fill the website lead form**
   Acts like a real company website

3. **Everything runs by itself**
   Lead saved → scored → assigned

4. **Open the Leads table**
   AI score and assigned salesperson

5. **Submit the same phone again**
   Matched to old lead, no duplicate

6. **Salesperson login → AI reply**
   Gemini writes, salesperson edits and approves

7. **Mark the lead Converted**
   Result saved to retrain the model

8. **Send the same webhook twice**
   Only one lead is created

Testing, Benchmarking & Performance

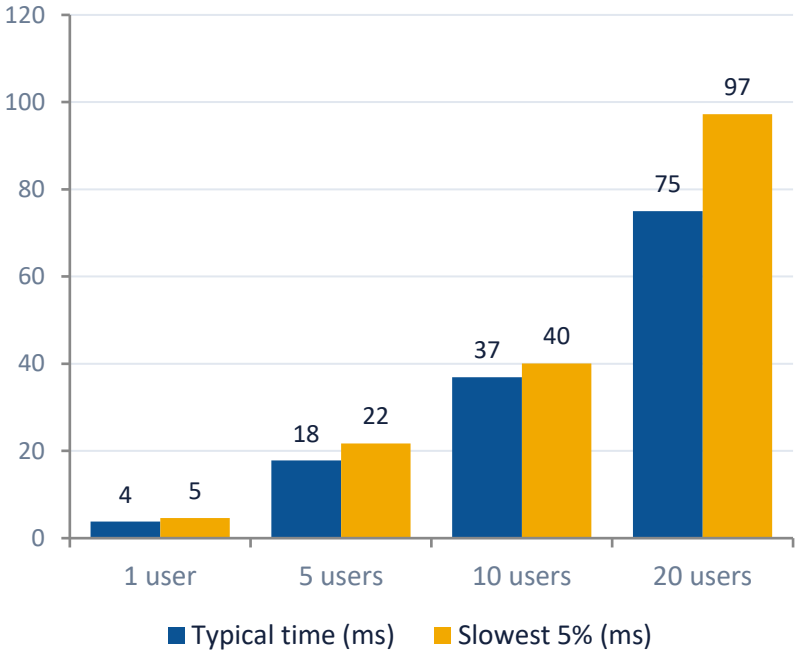
4.0 ms
time to score one lead (ML)

≈270 / s
leads scored per second, 1 worker

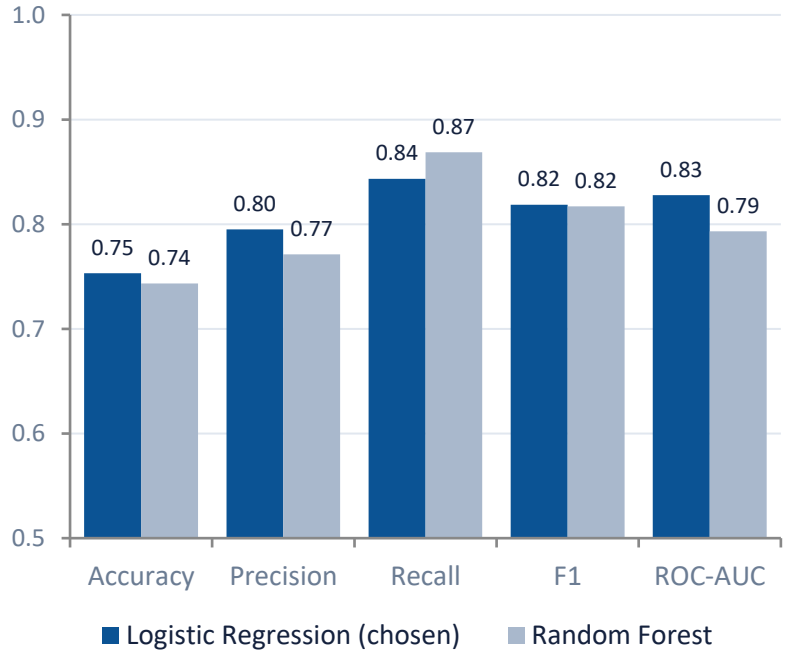
120 ms
to load the leads list (typical)

0.83
ROC-AUC (model ranking score)

ML scoring time vs users at once



Model comparison (300 test leads)



Backend response time

Request	Typical	Slowest 5%
Leads list	120 ms	180 ms
Dashboard numbers	438 ms	656 ms

1 user, server → MongoDB Atlas

Test scripts

Login & OTP · leads · master data · follow-ups, tasks, calendar · file upload · push alerts · full API test suite

Innovation, Domain Impact & Originality

What is new

From form to salesperson in seconds

Each new lead is checked for duplicates, scored by ML and given to the least busy salesperson. No manual sorting.

AI replies checked by a person

Gemini writes replies using only real company facts; the salesperson edits and approves before sending.

A model that learns from results

Every won or lost lead is saved, so the model can be retrained and get better.

Easy to add new lead sources

A new platform needs one small file; secure webhooks block fake and repeated leads.

Domain impact

For small and mid-size sales teams in education, real estate, cars and software:

- No two salespeople calling the same customer
- Work shared fairly across the team
- Best leads are called first
- Every action on a lead is recorded

Plagiarism check

[ADD] __ %
similarity index

[ADD] Tool & date
e.g. Turnitin / DrillBit

Attach the report to the project file

Cloud Deployment Specs & Scalability Roadmap

Component	Platform	Spec / configuration
Frontend	Vercel	React 19 + Vite build, served from a global CDN
Backend API	Render	Node.js 18+, Express 4 web service
ML service	Render	Python, FastAPI + Uvicorn, scikit-learn model
Database	MongoDB Atlas	Cloud MongoDB cluster, Mongoose 8
File storage	Cloudinary	Profile photos and lead attachments
Notifications	Firebase · SMS Hub India	App alerts, SMS and OTP login
Generative AI	Google Gemini API	Reply drafts from real company facts

Security: login tokens expire in 15 min (renewed for 7 days) · each role sees only its pages · max 300 requests per 15 min per IP (20 for login) · signed webhooks · unsafe input is cleaned

Scalability roadmap

NOW

- More backend copies can run side by side
- Scoring runs in the background
- ≈270 scores per second per ML worker

NEXT

- More ML workers and backend servers
- Job queue (Redis) for scoring and alerts
- Database indexes and auto-scaling

LATER

- Google Ads, LinkedIn, WhatsApp sources
- Retrain on real sales results
- Many companies on one system

IEEE References & Viva Defense

[1] P. Espadinha-Cruz, A. Fernandes, and A. Grilo, "Lead management optimization using data mining: A case in the telecommunications sector," Computers & Industrial Engineering, vol. 154, Art. no. 107122, 2021.

[2] A. A. Ohiomah et al., "The role of lead management systems in inside sales performance," Journal of Business Research, vol. 102, pp. 163–177, 2019.

[3] S. Das and B. Datta, "Application of UTAUT2 on adopting artificial intelligence powered lead management system (AI-LMS) in passenger car sales," Technological Forecasting and Social Change, vol. 201, Art. no. 123241, 2024.

[4] R. Nygård and J. Mezei, "Automating lead scoring with machine learning: An experimental study," in Proc. 53rd Hawaii Int. Conf. System Sciences (HICSS), 2020.

[5] M. Wu, P. Andreev, and M. Benyoucef, "The state of lead scoring models and their impact on sales performance," Information Technology and Management, vol. 25, no. 1, pp. 69–98, 2024.

[6] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," J. Mach. Learn. Res., vol. 12, pp. 2825–2830, 2011.

[7] L. Breiman, "Random forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[8] T. Fawcett, "An introduction to ROC analysis," Pattern Recognition Letters, vol. 27, no. 8, pp. 861–874, 2006.

Design decisions we defend

Why Logistic Regression?

Higher ROC-AUC (0.83 vs 0.79): it is better at putting likely buyers on top.

Why generated training data?

No real sales history yet; real results are added as they come in.

Why assign by workload?

It looks at how busy each salesperson is right now; taking turns does not.

Why must a person approve AI replies?

So wrong prices or false promises never reach a customer.